

# PPE detection on construction imagery

A recall-first safety screening model, and why it is not ready

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MAICEN — Module 4, Unit 3 (Computer Vision) · FMP group assignment

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## THE PROBLEM

# PPE compliance is checked by eye, intermittently

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### Today

A supervisor scans a site while doing several other things. Coverage is ad hoc and unrecorded.

### With a detector

Site photos and CCTV frames are pre-screened. The supervisor reviews a ranked worklist instead of scrolling through everything.

### Error priority

A missed no-helmet is a potential injury. A false alarm costs thirty seconds. We accept false positives to reduce misses.

Deployment is near real-time, not edge: 9 ms inference on a T4 is far faster than the human review loop around it.

## Five classes, one of them starved

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| Class            | Train instances | Meaning                                                    |
|------------------|-----------------|------------------------------------------------------------|
| person           | 7,075           | A person, regardless of PPE                                |
| helmet           | 6,330           | A hard hat, worn                                           |
| vest             | 3,221           | A high-visibility vest, worn                               |
| no-vest          | 2,222           | A torso with no high-visibility vest                       |
| <b>no-helmet</b> | <b>291</b>      | <b>A head with no hard hat — the safety-critical class</b> |

24 **person** instances for every 1 **no-helmet**. Roboflow-100 construction-safety-gsnvb, CC BY 4.0, 1,206 source images, native split kept for comparability with the published baseline.

## RESULTS

Strong overall. Weakest where it matters.

0.867

mAP@50

0.863

precision

0.787

recall, all

0.545

recall, no-helmet

The model clears the bar for a screening assistant and fails the bar it was built for. **no-helmet** precision is 0.947 — when it fires it is almost always right, but it fires too rarely. It behaves like a skeptic where the use case demands a paranoid.

mAP@50–95 of 0.474 against mAP@50 of 0.867: objects are found, boxes fit loosely. Fine for “someone here has no helmet”, not for counting.

| Class            | Precision    | Recall       | mAP@50       | mAP@50–95    | Val instances |
|------------------|--------------|--------------|--------------|--------------|---------------|
| person           | 0.864        | 0.919        | 0.945        | 0.611        | 246           |
| helmet           | 0.921        | 0.888        | 0.921        | 0.522        | 237           |
| vest             | 0.818        | 0.837        | 0.884        | 0.495        | 145           |
| no-vest          | 0.764        | 0.746        | 0.797        | 0.388        | 91            |
| <b>no-helmet</b> | <b>0.947</b> | <b>0.545</b> | <b>0.789</b> | <b>0.353</b> | <b>11</b>     |

The no-helmet row rests on 11 instances across 6 images — read it as a signal that the class is undertrained and under-validated, not as a precise measurement.

## Two findings that change the reading

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### 1 — The false positives are not errors

All three highest-confidence FPs are correct detections: a white hard hat at 0.91, two people at 0.88 and above. The labels are missing, not the objects.

Measured precision is therefore a **floor**, not a value. An audit of the validation labels has to precede any further training — improvements measured against noisy ground truth cannot be trusted.

The dataset is largely web-scraped studio and group photography, not site or CCTV frames — which is the deployment condition proposed.

### 2 — The misses are about context, not frequency

FN-1: a bare-headed figure in a white coat, missed entirely. FN-2: a soft cap, scored 0.21 — below threshold, not absent.

no-helmet appears to have learned site context and head-covering shape rather than the absence of a hard hat. More of the same stock imagery would not fix it.

# Four data actions, in order

| Action                                                                                 | Targets           | Effort |
|----------------------------------------------------------------------------------------|-------------------|--------|
| 1 Rebalance no-helmet — add 300–500 instances, prioritising caps, hoods and bare heads | FP-1, FP-2        | ~3 h   |
| 2 Expand validation support for rare classes to 50+ no-helmet instances                | measurement trust | ~1 h   |
| 3 Audit the validation labels                                                          | FP-1, FP-2        | ~2 h   |
| 4 Add hard negatives — buckets, cones, drums, bystanders                               | no-vest precision | ~2 h   |

**Explicitly not on the list:** more epochs, a larger backbone, heavier augmentation. Validation mAP plateaued at epoch 9 of 24. The ceiling is data coverage, not training.

# When this model must not be used

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## Limitations

Assistive screening only. Every detection is reviewed by a site supervisor before any action. Never the sole verifier for a life-safety decision.

Compliance cannot be assessed beyond roughly 30 px head height — a hard operating limit.

## Risk balance

A false negative is an unflagged injury risk. A false positive costs a supervisor thirty seconds. At current recall, roughly half of unhelmeted people pass unflagged. Stated plainly in the repository, not buried.

## Rights

Code MIT. Dataset CC BY 4.0, attribution retained, used unmodified. Weights inherit AGPL-3.0 from Ultralytics — distribution or networked deployment carries obligations. Faces visible in source imagery; no client site photography included.

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Repository · [github.com/caprijopi-alt/ppe-detection-yolov8](https://github.com/caprijopi-alt/ppe-detection-yolov8) · weights published as Release v1.0